

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

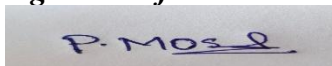
Scenario Based Assignment

Code of Conduct:

I P.Moshiya(192572142) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink that reads "P. Moshiya".

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic
- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. Explain how Hill Climbing would work in this scenario and identify its limitations
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search
- ii. Analyze the challenges of partial observability and dynamic environments
- iii. Describe how the rover builds and updates its internal model
- iv. Compare online search with uninformed and informed search strategies
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem
- ii. Analyze the computational challenges of large game trees
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv. Design an evaluation function and justify the factors considered
- v. Recommend a strategy for balancing optimality and real-time performance

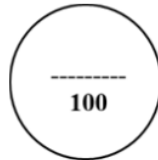
RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

CO2 – ASSESSMENT TOOL 2
Scenario Based Assignment

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17
QUESTIONS: 5 **Duration:** 1 Hour **Total Marks:** 50

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total (50)
Understanding of Scenario	20% (2)	/2	/2	/2	/2	/2	/10
Identification of Key Issues	20% (2)	/2	/2	/2	/2	/2	/10
Application of Concepts	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/2.5	/12.5
Decision Making	20% (2)	/2	/2	/2	/2	/2	/10
Clarity of Response	15% (1.5)	/1.5	/1.5	/1.5	/1.5	/1.5	/7.5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/50



Scenario 1: AI-Powered Drone for Emergency Medicine Delivery

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty.

Answer

Greedy Best-First Search (GBFS) selects the path that appears closest to the destination using the heuristic value $h(n)$ (straight-line distance).

Working

1. The drone starts from the source location.
2. It calculates the heuristic value (straight-line distance) for all neighboring locations.

3. It chooses the location with the smallest heuristic value.
4. It repeats this process until it reaches the destination.
5. If a path becomes blocked, the drone searches for another nearby path.

Strengths

- Uses heuristic information to reach the goal quickly.
- Explores fewer nodes than uninformed search.
- Suitable when fast decisions are required.
- Requires less computation time.
- Useful during emergency situations where speed is important.

Weaknesses

- Does not consider the actual travel cost.
- May choose unsafe or expensive routes.
- Can get trapped in a dead-end if the heuristic is misleading.
- Does not always produce the optimal path.
- Performance decreases when the environment changes frequently.

*ii. Explain how A would handle the same situation, including how it balances cost and heuristic.**

Answer

A* Search combines the actual travel cost and heuristic estimate.

It uses the evaluation function:

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$ = Actual cost from the start to the current node.
- $h(n)$ = Estimated cost from the current node to the goal.
- $f(n)$ = Total estimated cost.

Working

1. The drone starts at the source.
2. It calculates both the travel cost $g(n)$ and heuristic $h(n)$.

3. It computes $f(n)=g(n)+h(n)$.
4. It selects the node with the smallest $f(n)$ value.
5. If weather or terrain changes, $g(n)$ is updated.
6. If a road becomes blocked, A* recalculates the best alternative route.
7. The process continues until the goal is reached.

Advantages

- Finds the optimal route.
- Considers both distance and travel cost.
- Avoids unnecessary expensive paths.
- Handles changing travel costs better than Greedy Search.
- Produces safer and more reliable routes.

Disadvantages

- Requires more memory.
- Takes more computation time.
- Slower than Greedy Search in very large search spaces.

iii. Analyze the impact of heuristic accuracy on both algorithms.

Answer

The heuristic function estimates the remaining distance to the destination.

When the heuristic is accurate

- Greedy Search quickly reaches the destination.
- A* finds the optimal path efficiently.
- Fewer nodes are explored.
- Search time is reduced.
-

When the heuristic is inaccurate

- Greedy Search may choose incorrect routes.
- It may travel through blocked or costly paths.
- A* is less affected because it also considers the actual path cost.
- A* still has a high chance of finding the optimal solution.

Comparison

Greedy Best-First Search

Depends heavily on heuristic

Wrong heuristic leads to poor path

Faster

May not find optimal solution

A* Search

Uses heuristic and actual cost

More robust to heuristic errors

Slightly slower

Finds optimal solution if heuristic is admissible

iv. Discuss how dynamic changes (blocked paths) affect search performance.

Answer

In a flood-affected area, roads may become blocked at any time.

Effect on Greedy Best-First Search

- The selected route may suddenly become unavailable.
- The algorithm must restart or search for another nearby route.
- Previously explored nodes may become useless.
- Efficiency decreases.

Effect on A*

- Updates the actual travel cost.
- Recalculates a better alternative route.
- Adapts more effectively to changing conditions.
- Maintains better route quality.

Overall Impact

- More nodes may need to be explored.
- Search time increases.
- Computational cost increases.
- Frequent replanning becomes necessary.
- Reliable heuristic information becomes more important.

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety.

Answer

Recommended Algorithm: A* Search

Justification

1. A* considers both actual cost and heuristic distance.
2. It finds the shortest and safest route.
3. It adapts better to changing weather and blocked roads.
4. It avoids high-risk paths.
5. It provides optimal solutions when the heuristic is admissible.
6. It improves delivery reliability during emergencies.
7. Although it requires more computation than Greedy Search, the increase in safety and accuracy makes it the better choice.

Scenario 2: Smart City Traffic Signal Optimization (Hill Climbing & Simulated Annealing / Genetic Algorithm)

Problem Statement

A smart city uses an AI system to optimize traffic signal timings across hundreds of intersections. The objectives are to:

- Minimize total waiting time.
- Reduce fuel consumption.
- Decrease traffic congestion.
- Continuously adjust signal timings according to traffic flow.

However, the search space is very large, traffic conditions change frequently, and the system may get stuck in locally optimal solutions.

Formulate the problem as an optimization problem and explain why traditional search is inefficient.

Answer

Optimization Problem

The objective is to find the best traffic signal timing that minimizes:

- Total waiting time.
- Fuel consumption.
- Traffic congestion.

Each possible signal timing represents one solution (state).

The AI searches for the solution with the lowest cost.

Why Traditional Search is Inefficient

1. There are hundreds of traffic intersections.
2. Each intersection has many possible signal timings.
3. The number of possible combinations becomes extremely large.
4. Exploring every possible solution requires too much time.
5. Traffic conditions change continuously.
6. Traditional search cannot respond quickly to real-time traffic changes.
7. Therefore, optimization algorithms are more suitable.

i. Explain how Hill Climbing would work in this scenario and identify its limitations.

Answer

Hill Climbing Algorithm

Hill Climbing is a local search algorithm.

It starts with an initial traffic signal plan and continuously improves it.

Working

1. Start with the current traffic signal timing.
2. Measure its performance (waiting time, fuel usage, congestion).
3. Generate neighboring solutions by slightly changing signal timings.
4. Compare all neighboring solutions.
5. Select the solution with the best improvement.
6. Repeat the process until no better neighboring solution exists.

Advantages

- Simple to implement.
- Requires less memory.

- Produces quick solutions.
- Suitable for small optimization problems.
- Faster than exhaustive search.

Limitations

- Can get stuck at local maxima.
- Cannot always find the global optimum.
- Stops when no immediate improvement exists.
- Performs poorly in complex search spaces.
- Sensitive to the starting solution.

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation.

Answer

1. Local Maximum

A local maximum is a solution that is better than nearby solutions but not the best overall solution.

Real-World Example

The AI finds a traffic signal plan that reduces congestion at one junction but causes heavy traffic in nearby intersections.

The algorithm stops even though a better city-wide solution exists.

2. Plateau

A plateau is a flat area where many neighboring solutions have the same value.

Real-World Example

Several traffic signal timings produce nearly identical waiting times.

The algorithm cannot determine which direction is better.

Progress becomes very slow.

3. Ridge

A ridge is a narrow path where improvements require moving in multiple directions simultaneously.

Real-World Example

Reducing congestion requires changing signal timings at several nearby intersections together.

Changing only one signal does not improve traffic.

Hill Climbing struggles to move toward the better solution.

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations.

Answer

Simulated Annealing

Simulated Annealing is an improved version of Hill Climbing.

Working

1. Start with an initial solution.
2. Generate a neighboring solution.
3. If the new solution is better, accept it.
4. If it is worse, sometimes accept it with a certain probability.
5. Gradually reduce this probability over time.
6. Continue until the system stabilizes.

Advantages

- Escapes local maxima.
- Explores different areas of the search space.
- Finds better global solutions.
- Works well for dynamic optimization problems.

Genetic Algorithm

Genetic Algorithm is based on the principle of natural evolution.

Working

1. Generate a population of traffic signal plans.
2. Evaluate the fitness of each plan.
3. Select the best plans.
4. Apply crossover to combine good solutions.
5. Apply mutation to introduce small random changes.
6. Repeat the process until an optimal solution is found.

Advantages

- Searches many solutions simultaneously.
- Avoids local maxima.
- Produces high-quality solutions.

- Suitable for very large search spaces.
- Easily adapts to changing traffic conditions.

Comparison

Hill Climbing	Simulated Annealing	Genetic Algorithm
Uses one solution	Uses one solution	Uses many solutions
Can get stuck in local maxima	Escapes local maxima	Avoids local maxima effectively
Fast	Moderate speed	Moderate to high speed
May stop early	Explores more solutions	Explores many solutions simultaneously
Simple	More flexible	Best for complex optimization

iv. Recommend the best approach and justify your choice based on scalability and adaptability.

Answer

Recommended Algorithm: Genetic Algorithm

Justification

1. Handles very large search spaces efficiently.
2. Optimizes hundreds of traffic intersections simultaneously.
3. Produces near-optimal solutions.
4. Avoids local maxima through crossover and mutation.
5. Adapts quickly to changing traffic conditions.
6. Can optimize multiple objectives together.
7. Suitable for real-time smart city applications.
8. Provides better scalability than Hill Climbing.
9. More reliable for dynamic environments.

Scenario 3: Autonomous Rover on Mars (Online Search Agent)

Scenario

An autonomous rover is deployed on Mars to explore unknown terrain. It must:

- Navigate safely to collect samples.
- Avoid hazards such as craters and rocks.
- Operate with limited sensing capability.
- Make decisions without a complete map.
- Update its knowledge as it explores.

i. Explain why this problem requires an online search agent instead of offline search.

Answer

An **online search agent** makes decisions while exploring the environment.

Explanation

1. The rover does not have a complete map of Mars.
2. It cannot calculate the entire path before starting.
3. It discovers new obstacles while moving.
4. It must make decisions based on current observations.
5. It continuously updates its path during exploration.
6. Therefore, an online search agent is more suitable than an offline search agent.

Offline Search vs Online Search

Offline Search

Complete map is known before starting.

Entire path is planned in advance.

Cannot easily adapt to new obstacles.

Suitable for static environments.

Online Search

Map is unknown initially.

Path is planned during movement.

Adapts to newly discovered obstacles.

Suitable for dynamic and unknown environments.

ii. Analyze the challenges of partial observability and dynamic environments.

Answer

Partial Observability

Partial observability means the rover can see only a limited area around it.

Challenges

1. The rover cannot detect distant obstacles.
2. Hidden craters may suddenly appear.

3. The rover has incomplete information.
4. It may choose an unsafe path.
5. Additional sensing is required before moving.

Dynamic Environment

A dynamic environment changes while the rover is operating.

Challenges

1. Dust storms reduce visibility.
2. Loose rocks may block the planned path.
3. Terrain conditions may change unexpectedly.
4. Previously safe routes may become dangerous.
5. The rover must continuously re-plan its path.

iii. Describe how the rover builds and updates its internal model.

Answer

The rover maintains an **internal model** of its environment.

Working

1. The rover starts with little or no map.
2. It collects information using cameras and sensors.
3. It stores information about rocks, craters, and safe paths.
4. It updates the map after every movement.
5. New obstacles are added to the internal model.
6. Unsafe areas are marked as blocked.
7. Safe routes are recorded for future navigation.
8. The internal model becomes more accurate over time.

Advantages

- Improves navigation.
- Reduces repeated exploration.
- Helps avoid hazards.
- Supports better decision-making.

iv. Compare online search with uninformed and informed search strategies.

Answer

Feature	Online Search	Uninformed Search	Informed Search
Environment	Unknown	Usually known	Usually known
Uses heuristic	Sometimes	No	Yes
Complete map required	No	Yes	Yes
Decision making	During exploration	Before exploration	Before exploration
Adaptability	High	Low	Moderate
Suitable for Mars rover	Yes	No	Limited

Explanation

Online Search

- Learns while moving.
- Updates knowledge continuously.
- Best for unknown environments.

Uninformed Search

- Uses no heuristic.
- Explores many unnecessary paths.
- Slow in large environments.

Informed Search

- Uses heuristic information.
- Faster than uninformed search.
- Requires more knowledge of the environment.

v. Justify the most suitable approach considering uncertainty, adaptability, and safety.

Answer

Recommended Approach: Online Search Agent

Justification

1. The Mars environment is unknown.
2. The rover has only partial information.
3. New obstacles appear during exploration.
4. The rover must adapt its decisions in real time.
5. Online search updates the path continuously.
6. It reduces the risk of collisions.

7. It improves navigation safety.
8. It increases mission success.
9. It works efficiently under uncertainty.

Scenario 4: University Exam Timetable (Constraint Satisfaction Problem - CSP)

Scenario

A university is designing an AI system to automatically generate exam timetables. The system must satisfy the following constraints:

- No student should have overlapping exams.
- Limited number of exam halls.
- Certain subjects must be scheduled before others.
- Faculty availability constraints.
- Special accommodations for some students.
- The complexity increases as the number of courses and students grows.

Formulate the problem as a Constraint Satisfaction Problem (CSP).

Answer

A **Constraint Satisfaction Problem (CSP)** consists of **Variables, Domains, and Constraints**.

- **Variables:** The exams that need to be scheduled.
- **Domains:** Available dates, time slots, and exam halls for each exam.
- **Constraints:** Rules that must be satisfied while creating the timetable.

The objective is to assign a valid time slot and exam hall to every exam without violating any constraints.

i. Clearly define variables, domains, and constraints with explanation.

Answer

1. Variables

Variables represent the exams to be scheduled.

Examples:

- Mathematics Exam
- Physics Exam
- Chemistry Exam

- Computer Science Exam

Each subject is treated as one variable.

2. Domains

The domain is the set of possible values that can be assigned to each variable.

Examples:

- Time Slot 1 (9 AM – 12 PM)
- Time Slot 2 (1 PM – 4 PM)
- Time Slot 3 (9 AM – 12 PM Next Day)

Exam halls:

- Hall A
- Hall B
- Hall C

Each exam can be assigned one available time slot and one exam hall.

3. Constraints

The timetable must satisfy the following constraints:

1. No student should have two exams at the same time.
2. One exam hall cannot conduct two exams simultaneously.
3. Faculty members must be available during the exam.
4. Some subjects must be scheduled before others.
5. Students requiring special accommodations must receive appropriate arrangements.
6. All exams must be assigned exactly one valid time slot and one hall.

ii. Explain how backtracking search works in this scenario.

Answer

Backtracking is a search algorithm used to solve CSPs by assigning values step by step.

Working

1. Select the first exam.
2. Assign an available time slot and exam hall.
3. Check whether all constraints are satisfied.
4. If the assignment is valid, proceed to the next exam.
5. If a constraint is violated, undo the assignment.
6. Try another available time slot or hall.

7. Repeat until all exams are successfully scheduled.

Advantages

- Finds a valid timetable.
- Ensures all constraints are satisfied.
- Simple and systematic approach.

Disadvantages

- Slow for very large timetables.
- May require many backtracking steps.

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency.

Answer

Constraint propagation reduces the search space by removing invalid choices early.

Forward Checking

Forward checking checks future assignments before making them.

Working

1. Assign a time slot to an exam.
2. Remove conflicting time slots from the domains of remaining exams.
3. If any exam has no remaining valid time slot, backtrack immediately.
4. Otherwise, continue assigning the next exam.

Advantages

- Detects conflicts early.
- Reduces unnecessary searching.
- Decreases the number of backtracking steps.
- Improves execution speed.
- Produces the timetable more efficiently.

iv. Analyze real-world challenges and justify the best strategy for scalability.

Answer

Real-World Challenges

1. Large number of students.
2. Hundreds of exams.
3. Limited exam halls.
4. Faculty availability changes.
5. Last-minute timetable modifications.

6. Special accommodation requirements.
7. Multiple departments sharing resources.

Best Strategy

The best strategy is **Backtracking Search with Forward Checking (Constraint Propagation)**.

Justification

1. Handles multiple constraints efficiently.
2. Detects conflicts early.
3. Reduces unnecessary searches.
4. Produces valid timetables faster.
5. Easily scales for large universities.
6. Improves overall scheduling performance.

Scenario 5: Strategic Game AI (Minimax & Alpha-Beta Pruning)

Scenario

A gaming company is developing an AI opponent for a real-time strategy game. The AI must:

- Compete against human players.
- Make decisions within strict time limits.
- Evaluate complex game states.
- Handle a very large decision tree.

i. Explain how the Minimax algorithm models this problem.

Answer

The **Minimax algorithm** is used for two-player games where one player tries to maximize the score and the other tries to minimize it.

Working

1. The AI considers all possible moves.
2. It assumes the opponent will always make the best move.
3. MAX player tries to maximize the score.

4. MIN player tries to minimize the score.
5. The algorithm evaluates the game tree.
6. It selects the move with the best possible outcome.

Advantages

- Makes rational decisions.
- Considers future moves.
- Produces optimal decisions when the search tree is complete.

ii. Analyze the computational challenges of large game trees.

Answer

Challenges

1. Every move creates many new possibilities.
2. The game tree grows exponentially.
3. Millions of game states may need evaluation.
4. Memory usage becomes very high.
5. Processing time increases significantly.
6. Real-time games require quick decisions.
7. Searching the complete tree is often impossible.

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning.

Answer

Alpha-Beta Pruning improves Minimax by eliminating branches that cannot affect the final decision.

Working

1. Perform Minimax search.
2. Maintain two values:
 - **Alpha (α):** Best value found for the MAX player.
 - **Beta (β):** Best value found for the MIN player.
3. If **Alpha $\geq$ Beta**, stop exploring that branch.
4. Ignore branches that cannot improve the result.
5. Continue evaluating only useful branches.

Advantages

- Reduces the number of nodes explored.

- Saves computation time.
- Uses less memory.
- Allows deeper game-tree search.
- Produces the same optimal result as Minimax.

iv. Design an evaluation function and justify the factors considered.

Answer

An **evaluation function** estimates the quality of a game position.

Example Evaluation Function

Evaluation Score = Material + Position + Mobility + King Safety

Factors Considered

- 1. Material Advantage**
 - Number and strength of units.
 - More powerful units increase the score.
- 2. Position**
 - Control of important areas.
 - Better positioning increases advantage.
- 3. Mobility**
 - Number of legal moves available.
 - More movement options improve flexibility.
- 4. Safety**
 - Protection of important units or base.
 - Safer positions receive higher scores.
- 5. Attack Opportunities**
 - Ability to capture opponent units.
 - Strong attacks increase the evaluation score.

Justification

These factors help the AI choose moves that improve its overall position and increase the chances of winning.

v. Recommend a strategy for balancing optimality and real-time performance.

Answer

Recommended Strategy

Use **Minimax with Alpha-Beta Pruning** and an **Evaluation Function**.

Justification

1. Alpha-Beta Pruning reduces unnecessary searches.
2. The AI evaluates only important branches.
3. Decision-making becomes much faster.
4. The evaluation function estimates good moves without searching the entire tree.
5. Suitable for real-time strategy games.
6. Produces strong and near-optimal decisions.
7. Balances accuracy and computation time effectively.